

Reference-Defined Exact-Instance Recognition

Route A / new HYP: Mechanism, Evidence, and Next Steps

Research Progress Report

September 12, 2026

Experimental evidence through September 11, 2026

One-sentence summary

Reference-conditioned matching and content evidence enable retrieval-supervised correction of a frozen retriever. Internal gains are established; complete spatial localization and independent external reliability are not.

1. Objective: recognize the exact instance, not just its category

The practical task

- Input: a cluttered natural query image and a full gallery of visual references.
- Output: the exact drug-package identity, not merely “a medicine box.”
- Challenges: near-duplicate packaging, partial visibility, distractors, and incorrect initial retrieval.

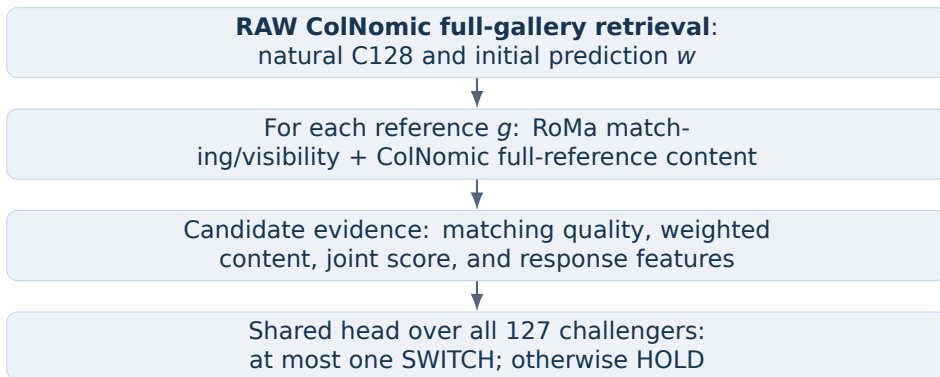
Long-term goal and constraints

- Enroll a new identity by adding a reference, without identity-specific retraining.
- Task supervision: query-reference identity pairs; no additional masks, boxes, or points.
- Learn shared comparison rules, not identity-specific classifiers.

Current milestone: test whether reference-conditioned evidence improves target-free recognition and correction.

Backbone pretraining supervision must be disclosed separately. Shared parameters alone do not establish enrollment reliability.

2. Current system: retain RAW, correct with reference evidence



RAW means unadapted ColNomic retrieval, not a raw image file format.
Reference image content affects scores; identity labels are used for indexing, supervision, and output.

All positive results here use RAW natural C128, not D1. The effective pipeline does not require superregions or a separate P/V module.

3. Mechanism: matching quality and identity content interact

For candidate g , let u_i^g, v_j^g be RoMa query/reference visibility weights, and c_{ij}^g be ColNomic token-content similarity.

$$\ell_i^g = \max_{j \in R_g} [v_j^g c_{ij}^g],$$

$$L_g = \frac{\sum_i u_i^g \ell_i^g}{\sum_i u_i^g},$$

$$M_g = \sqrt{\text{mean}_i(u_i^g) \text{mean}_j(v_j^g)},$$

$$S_g = M_g L_g.$$

- **RoMa** supplies cross-image matching/visibility weights, not a standalone identity decision.
- **ColNomic** searches the complete reference, rather than reading only one hard correspondence.
- A **shared head** combines RAW gap and relative quality, content, and response evidence.

Computational structure is not a reliability guarantee

These idealized real-valued formulas assume valid denominators; the implementation retains numerical safeguards and its frozen FP64 operation order. Strong matching need not imply the correct identity.

NEW_HYP_THEORY_DEFINITION_AND_PROPOSITIONS_V2: definitions, information interfaces, and product interaction.

4. new HYP: select the most credible identity explanation

What is the hypothesis?

“Candidate reference g explains the query identity,” not “this connected region is the target.” This formalizes the effective mechanism; it is not a newly validated spatial hypothesis network.

Fix the RAW winner w . The shared head outputs z_g ; HOLD has score zero:

$$g^* = \arg \max_{g \in C_{128} \setminus \{w\}} z_g, \quad \hat{y} = \begin{cases} g^*, & z_{g^*} > 0, \\ w, & z_{g^*} \leq 0. \end{cases}$$

An ideal action-value target is $\Delta_g(Z) = P(Y = g | Z) - P(Y = w | Z)$.

- The correct challenger must beat **both HOLD and every incorrect challenger**.
- Training uses identity relationships; inference does not know the correct candidate.
- z_g is a surrogate score, **not a proven calibrated probability difference**. HOLD is not unknown-class rejection.

Unified working hypothesis V3. Defining the ideal target does not prove consistency of the implemented surrogate loss.

5. What has been addressed?

Problem	Current progress	Boundary
Initial retrieval confuses near duplicates	Reference-conditioned quality and content evidence supplement RAW	Internal gains, not universal correctness
Useful signals do not produce actions	Full-C128, 127-challenger HOLD/SWITCH is implemented	Not merely a target-known pair diagnostic
An extra module might only add a global bias	Binding interventions remove original rescues; the increment has a matched bias control	Computational dependence, not pixel causality
More training images can reduce accuracy	On identical ALL rows, group weighting improves 440→447	A protocol-specific factor, not a universal root cause

We now have an

implemented, empirically supported reference-conditioned correction mechanism.

6. Main results: each model is compared with its own RAW baseline

Model	Evaluation	RAW	Model	Rescue / break	Net
FROZEN_C	difficult90	61/90	69/90	8/0	+8
ORIGINAL7	Original EVAL32	25/32	28/32	3/0	+3
ORIGINAL7	Additional EVAL128	88/128	99/128	12/1	+11
ALL_COND	593-query OOF5	426/593	445/593	19/0	+19
GROUP_BASE	Same 593-query OOF5	426/593	447/593	22/1	+21

Expanded internal evidence

593 queries, 68 identities, and 64 identity/source connected groups; each fold excludes its held-out groups from training. GROUP_BASE improves RAW from 71.84% to 75.38%: +3.54 percentage points.

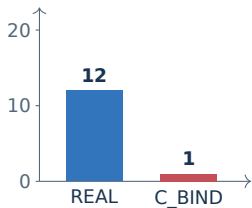
Do not pool these rows: FROZEN_C and ORIGINAL7 use different parameters. OOF5 refits per fold; it is not one frozen head evaluated on a growing test set. All results are internal/development evidence, not untouched external confirmation.

Original JSON and independent-review paths are listed in README. Denominators count queries, not candidate width.

7. Mechanistic evidence: does the correct binding matter?

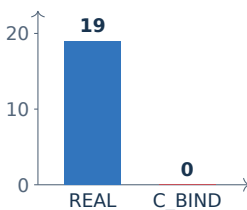
The vertical axis is **retention of REAL's original rescue set**, not total control accuracy.

ORIGINAL7 / EVAL128



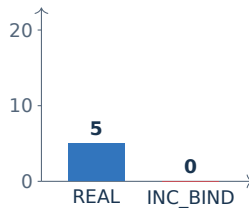
Rebind the full evidence bundle
Total correct: 99→67

ALL_COND / 593 OOF



Rebind the full evidence bundle
Total correct: 445→397

Increment / 593 OOF



Freeze BASE; rebind only the increment
Total correct: 445→441

The observed corrections depend on evidence bound to the specific reference.

A matched bias-only increment stays at 440 and does not reproduce the five added rescues. The third analysis is post-result exploratory attribution, not proof of token correspondence or pixel causality.

ORIGINAL7 EVAL128; H593 positive retrieval and increment attribution closure.

8. Optimization: one effective correction, not evidence for more modules

Same data and model; change group weighting

440 $\longrightarrow$ 447

8 rescues / 1 break; net +7
Equal-group difference: +1.218 pp;
exact group test $p = 0.0078125$.

The same seven-parameter architecture; no new backbone or task spatial labels.

Stronger comparisons remain limited

- SMALL_CONST (446) vs. GROUP_BASE (447): 4 rescues / 3 breaks; no robust advantage established.
- GROUP_BASE (447) vs. GROUP_COND (446): the extra conditional gate is not shown necessary.
- The predefined GROUP_COND vs. ALL_COND primary comparison did not establish a robust advantage.

Interpretation: training-population weights affect transfer, but a positive component comparison does not turn every experimental gate into a pass.

Unified hypothesis prediction readout, 2026-09-11. Reweighting the old 32/128 protocol did not improve the original 28/99.

9. Remaining bottleneck: inclusion is not correct selection

GROUP_BASE: 593-query OOF5

Correct final identity	447
Target in C128, final answer wrong	123
Target absent from C128	23
Total	593

C128 coverage: $570/593 = 96.12\%$.

The last three failure locations have appeared in prior diagnostics; we do not assign a single common cause to all 123 in-set errors.

593 per-query ledger; ORIGINAL28/99 plateau established boundaries.

Not simply “there is no evidence”

- Some targets are missing from the candidate set.
- An incorrect candidate may have stronger joint evidence.
- A target may beat other challengers but fail to beat HOLD.
- More aggressive switching may damage correct RAW predictions.

10. new HYP and spatial HYP are different claims

Current new HYP

Candidate identity explanation and decision.

- A reference organizes quality, content, and competition evidence.
- Internal correction and candidate-binding evidence exist.
- Connected segmentation is not a prerequisite.

A scope change is not proof of the old claim: spatial localization, separate P/V closure, and pixel-level ownership still require their own evidence.

RoMa-RGH S0 job 5139368 final audit; new HYP definitions V2 and unified working hypothesis V3.

Earlier spatial RGH/HYP

Reference-generated connected target regions.

- S0: legal target H1 in 31/32, but primary ranking only 7/32.
- Candidate-dependent counts of legal regions biased the score.
- This fixed family was NO-GO; it does not overwrite the effective correction results.

11. Next: test unseen identities without refitting the test images

1. **Freeze the original 32-query experiment's model.** No encoder, head, or threshold updates.
2. **Audit actual training identities.** Select identities among the 593 queries absent from this checkpoint's task training, not merely absent from its original 32 test queries.
3. **Keep their references; test other natural queries.** Full-gallery retrieval and natural C128, with no target insertion.
4. **Report the complete population.** Previously successful cases may be a diagnostic slice; evaluate all queries of the selected identities for recall, final accuracy, rescues, and breaks.

Two distinct capabilities

Unseen-identity testing measures transfer of the shared model. Dynamic enrollment additionally tests old-identity retention before and after new references are added.

Planned, not completed. Audit reference exposure as training negatives separately; reused development data are not untouched external confirmation.

12. Take-home message

What we have achieved

We implemented reference-conditioned exact-instance correction with no additional task spatial labels. Shared retrieval supervision learns how matching and content evidence support switching or retaining the original prediction over a complete natural C128.

- **Performance:** net +11 on 128 additional queries; 19/0 rescues/breaks for the conditional 593-query method and 22/1 for the group-weighted method.
- **Mechanism:** rebinding candidate evidence removes original rescues, supporting its specific-reference role.
- **Theory:** explicit evidence decomposition, action objectives, and testable hypotheses; established mathematical properties are not claimed as new theorems.
- **Limits:** independent external generalization, complete spatial localization, and universal no-regret behavior remain unproven.

Positioning: reproducible internal gains with mechanistic support; next, test transfer and reliability.